

Exact empirical quadratic residual queries: sufficient moments and qualified memory lower bounds

Phase7 research note, memory route

7 October 2026

Decision. There is a rigorous, narrow mathematical route: characterize the information needed to answer every empirical residual query after history has been discarded. The canonical number of real coordinates is

$$p = \binom{d+1}{2}, \quad q = \binom{d+3}{4}, \quad D_0 = p + q, \quad D = 2p + q.$$

Here D_0 counts input moments of orders two and four, and D also counts the label-quadratic moment. For fixed unweighted sample size $n \geq D_0$, D is the exact minimum dimension among continuous real-valued summaries that work on every dataset. If the label moment is already supplied, the exact additional continuous dimension is D_0 . A local finite-bit packing bound is also available, with its accuracy metric and domain specified below. These are specializations of classical normal-equation, moment-space, linear-algebra and topological arguments. They do not establish an information-theoretic advantage of the existing learner, an optimal training method, or a Gaussian average-case memory bound.

Data / guarantee	Memory model	Valid conclusion
Every size- n unweighted dataset; $n \geq D_0$	Continuous encoder; arbitrary decoder	Exactly D real coordinates; D_0 extra given S_y .
Every weighted finite probability measure	Additive real measurements; arbitrary decoder	Exactly D independent measurements.
Every size- n unweighted dataset, any n	Additive encoder; affine query decoding	At least D measurements.
Every size- n unweighted dataset, small n	Continuous encoder; arbitrary decoder	Lower bound r_n ; raw data give upper bound $n(d+1)$.
Every dataset in a specified local compact family; error on (4)	At most 2^b discrete states	Local packing bound (7), with specified r, ε .
Equal-weight datasets; no regularity	Infinite-precision additive real state	One real can encode the full finite multiset.

1 Definition, query equivalence and a matching summary

Fix $d, n \geq 1$ and a known sample count n . Data are an ordered tuple $z = ((x_i, y_i))_{i=1}^n \in (\mathbb{R}^d \times \mathbb{R})^n$. Permutation invariance is not required of a competing summary; allowing order only strengthens

the lower bound. Set

$$H(x) = xx^\top - I_d, \quad q_B(x) = \text{tr}(BH(x)),$$

$$G_z(B, c) = \frac{1}{2n} \sum_{i=1}^n (y_i - q_B(x_i) - c)H(x_i), \quad B \in \text{Sym}_d, \quad c \in \mathbb{R}.$$

A summary $S_n : z \mapsto \mathbb{R}^m$ is *universally query-sufficient* if there is a decoder Γ_n such that

$$\forall z \forall B \in \text{Sym}_d \forall c \in \mathbb{R} : \quad \Gamma_n(S_n(z), B, c) = G_z(B, c). \quad (\text{U})$$

No regularity of Γ_n is needed for the continuous lower bound. The decoder has no additional data-dependent side information unless explicitly stated; all retained model/optimizer/replay information that it accesses must be counted in the summary. A known sample count is free in this fixed- n model. A streaming implementation stores the count too.

Write $\langle a \rangle_z = n^{-1} \sum_i a(x_i, y_i)$ and define

$$S_y = \frac{1}{2} \langle yH \rangle_z, \quad M_{2,ij} = \langle x_i x_j \rangle_z, \quad M_{4,ijkl} = \langle x_i x_j x_k x_l \rangle_z.$$

The subscripts on M_2, M_4 are coordinate indices rather than sample indices. Store only distinct symmetric coordinates: $i \leq j$ for M_2 and $i \leq j \leq k \leq l$ for M_4 . Let

$$F_n(z) = M(z) = (S_y, M_2, M_4) \in \mathbb{R}^D.$$

The order-four tensor has q distinct entries, not $p(p+1)/2$ independent entries. Products of centered quadratic features have polynomial relations, and known total mass is not an additional unknown coordinate.

Theorem 1 (Canonical query information). *For any two datasets z, z' of the same known size, the following are equivalent:*

1. $G_z(B, c) = G_{z'}(B, c)$ for every symmetric B and every c ;
2. $(S_y, \overline{H}, T)_z = (S_y, \overline{H}, T)_{z'}$, where $\overline{H} = \langle H \rangle$ and $T(B) = \frac{1}{2} \langle \langle B, H \rangle H \rangle$;
3. $M(z) = M(z')$.

Consequently M answers every query, and every sufficient summary permits recovery of M . This is minimality for a deterministic query family, not a statement of statistical sufficiency for an unknown parameter.

Proof. Direct expansion gives

$$G_z(B, c) = S_y - T(B) - \frac{1}{2} c \overline{H}. \quad (1)$$

Thus the value at $(0, 0)$ determines S_y , its difference from $(0, 1)$ determines $\overline{H}$, and its differences from queries at a basis of Sym_d determine T . Conversely these coefficients determine every query.

For $k \leq l$, put $F^{kk} = e_k e_k^\top$ and $F^{kl} = (e_k e_l^\top + e_l e_k^\top)/2$ if $k < l$, so that $\langle F^{kl}, H \rangle = H_{kl}$. Define

$$K_{ij,kl} = \langle H_{ij} H_{kl} \rangle = 2[T(F^{kl})]_{ij}.$$

The reconstruction formulas are

$$M_{2,ij} = \overline{H}_{ij} + \delta_{ij}, \quad (2)$$

$$M_{4,ijkl} = K_{ij,kl} + \delta_{ij} M_{2,kl} + \delta_{kl} M_{2,ij} - \delta_{ij} \delta_{kl}. \quad (3)$$

Conversely (3) reconstructs K from M_2, M_4 , and hence reconstructs T . All index permutations are determined by tensor symmetry. If two histories have the same sufficient summary, (U) makes all their queries agree, so the reconstructed M agrees too. Therefore a map h on the summary image satisfies $M = h \circ S_n$. $\square$

The $p + 2$ fixed queries

$$\mathcal{Q} = \{(0, 0), (0, 1)\} \cup \{(F^{kl}, 0) : 1 \leq k \leq l \leq d\} \quad (4)$$

already recover all coefficients. This reduces any exact universal adaptive-query claim to a finite nonadaptive query family. Once M is stored, correctness is pathwise and holds for arbitrarily data-dependent choices of B, c ; independence or generalization assumptions are unnecessary. If only one prespecified algorithm trajectory must be supported, (U) is a stronger requirement and the lower bound need not apply.

The summary is additive: store sums of $yH/2$, all degree-two input monomials, and all degree-four input monomials, then divide by n . This uses D ideal real coordinates and a count, independently of stream length. It is an algebraic exact-arithmetic representation; ordinary floating-point accumulation has rounding error. Nothing here identifies the real-coordinate count with a physical byte count.

2 Continuous summaries on fixed unweighted samples

Let $f(x) \in \mathbb{R}^{D_0}$ list all degree-two and degree-four monomials. The functions $1, f_1, \dots, f_{D_0}$ are linearly independent. The additional p functions $yH_{ij}(x)/2$ are independent modulo this input-only span. Indeed, the coefficient of y in a linear relation would say $x^\top Ax - \text{tr}(A) \equiv 0$, forcing $A = 0$; independence of different input monomials then finishes the argument. Thus the affine span of the single-example canonical feature vector is all of $\mathbb{R}^D$.

Lemma 2 (A full-dimensional unweighted moment image). *For every fixed $n \geq D_0$, there is a dataset z_0 at which the polynomial moment map $F_n : (\mathbb{R}^{d+1})^n \rightarrow \mathbb{R}^D$ has differential of rank D . Consequently its image contains a nonempty open subset of $\mathbb{R}^D$, and there is a continuous local section $s : U \rightarrow (\mathbb{R}^{d+1})^n$ on some nonempty open $U \subset \mathbb{R}^D$ such that $F_n(s(u)) = u$.*

Proof. First consider $A_n(x_1, \dots, x_n) = n^{-1} \sum_i f(x_i)$. The vectors $\partial_a f(x)$, over all $x \in \mathbb{R}^d$ and coordinates a , span $\mathbb{R}^{D_0}$. Otherwise a nonzero v annihilates them all, so every partial derivative of the polynomial $v^\top f$ vanishes. That polynomial is constant on $\mathbb{R}^d$, contradicting independence of $1, f_1, \dots, f_{D_0}$.

Select D_0 such derivative vectors forming a basis, and use their locations as D_0 sample points. Their selected derivative columns in DA_{D_0} are independent. Pad the configuration when $n > D_0$. Hence some D_0 -minor of DA_n is a nonzero polynomial in the sample coordinates. The configurations where it is nonzero form an open dense set $\mathcal{U}$.

Independence of the p polynomials H_{ij} implies that there are p points for which the $p \times p$ evaluation matrix has rank p : if their value vectors never spanned $\mathbb{R}^p$, a nontrivial linear combination of the functions would vanish everywhere. Since $n \geq D_0 \geq p$, the $p \times n$ matrix with columns $H(x_i)$ in symmetric coordinates has rank p on another open dense set $\mathcal{V}$. Both density assertions follow because a nonzero real polynomial cannot vanish on an open set. The finite intersection $\mathcal{U} \cap \mathcal{V}$ is nonempty.

Choose an input configuration there and set every label to zero. At this dataset, the derivative of S_y with respect to input coordinates is zero, while its label-derivative columns are $H(x_i)/(2n)$,

of rank p . The derivative of (M_2, M_4) with respect to inputs has rank D_0 and its label derivatives vanish. Thus the full differential has rank $p + D_0 = D$. Choose D independent scalar input/label coordinate directions and freeze the others. The inverse function theorem applied to the resulting square map supplies U and its smooth, hence continuous, local section. $\square$

Theorem 3 (Tight continuous dimension for sufficiently many samples). *Fix $n \geq D_0$. Any continuous universally query-sufficient summary $S_n : (\mathbb{R}^{d+1})^n \rightarrow \mathbb{R}^m$ must have $m \geq D$. This bound is attained by M . If S_y is supplied separately and an additional continuous summary A_n has values in $\mathbb{R}^m$, exact universal decoding from (S_y, A_n) requires and permits $m \geq D_0$.*

Proof. On the local section in the lemma, $S_n \circ s : U \rightarrow \mathbb{R}^m$ is continuous. It is injective: equal summary values imply equal query coefficients by the first theorem, and $F_n(s(u)) = u$. If $m < D$, compose this injection with the usual coordinate embedding $\mathbb{R}^m \hookrightarrow \mathbb{R}^D$. Brouwer’s invariance of domain says its image is open in $\mathbb{R}^D$, but it lies in an m -dimensional coordinate subspace with empty interior. This is a contradiction.

For separate S_y , the combined map (S_y, A_n) is a continuous summary of dimension $p + m$, so $p + m \geq D$. Storing M_2, M_4 attains $m = D_0$. Neither proof assumes continuity of the decoder. $\square$

Finite sample qualification. The sufficient threshold $n \geq D_0$ is not claimed to be the smallest threshold. For general n , let

$$r_n = \max_z \text{rank } DF_n(z).$$

The same independent-minor argument and a local parameter slice imply that every continuous universal summary has dimension at least r_n . Moreover $r_n \leq \min\{D, n(d+1)\}$ and storing the ordered raw data uses $n(d+1)$ continuous coordinates. Thus a D lower bound cannot hold when $n(d+1) < D$. At $n = 1$, for example, storing (x, y) uses only $d+1$ coordinates. These bounds do not determine the exact minimum for every small n ; a sharper small- n classification can involve the geometry of finite-atomic polynomial moment varieties.

Quantifiers and model restrictions. The theorem assumes correctness at every dataset in the stated unrestricted domain. It is not automatically a lower bound for data on a fixed noiseless teacher manifold, for a discrete alphabet, or for almost-sure correctness with an arbitrary discontinuous decoder. Even topological full support of a sampling distribution does not turn an almost-sure factorization through a discontinuous decoder into a pointwise factorization. If both encoder and the finitely many decoders in (4) are continuous, equality almost surely under a full-support law extends by continuity to every dataset, so the theorem then applies. A fixed shared random seed also reduces to this deterministic theorem only if that seed’s map has universal correctness.

3 Linear measurements and why additivity alone is insufficient

The domain in this section is all finitely supported probability measures μ on $\mathbb{R}^{d+1}$, with arbitrary nonnegative real weights, not merely equal weights. Let $\phi_1, \dots, \phi_m$ be any real-valued functions and store their linear measurements $L(\mu) = (\int \phi_j d\mu)_j$. Here “linear” means linear in the measure, not linear in x or y .

Proposition 4 (Exact linear-measurement criterion). *For a target function f , the scalar $\int f d\mu$ is determined by $L(\mu)$ for every finitely supported probability measure if and only if*

$$f \in \text{span}_{\mathbb{R}}\{1, \phi_1, \dots, \phi_m\}. \tag{5}$$

Therefore universal exact quadratic residual queries require $m \geq D$ in this measurement model, regardless of decoder regularity. The canonical measurements attain equality.

Proof. The “if” implication follows by integrating a linear representation, using known total mass one. For the converse, put $v(z) = (1, \phi_1(z), \dots, \phi_m(z))$. If every finite real relation $\sum_i a_i v(z_i) = 0$ also implied $\sum_i a_i f(z_i) = 0$, then $\ell(\sum_i a_i v(z_i)) = \sum_i a_i f(z_i)$ would be a well-defined linear functional on the span of the $v(z)$. Extend it to $\mathbb{R}^{m+1}$; then $f(z) = \ell(v(z))$, establishing (5). If (5) fails, there is instead a finite relation with $\sum_i a_i = 0$ and $\sum_i a_i f(z_i) \neq 0$. Split the coefficients into positive and negative parts. Their common positive mass a normalizes them into two finitely supported probability measures μ_+, μ_- with equal L but unequal integrals of f . No decoder can distinguish them.

Apply this criterion to the D canonical feature functions. Their classes modulo constants are linearly independent, as proved above, so the span of m measurement classes must have dimension at least D . $\square$

An equivalent finite-support kernel statement is $\ker L \cap \ker(\text{mass}) \subseteq \ker M$: every signed measure annihilated by the stored measurements and mass must annihilate all query coefficients. This is ordinary linear algebra. It is not a separate streaming or information-theoretic principle.

Two models that must not be conflated. For equal-weight samples of a fixed size, arbitrary discontinuous additive features do not satisfy the lower bound in the proposition. Choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$ and assign each possible record $z \in \mathbb{R}^{d+1}$ to a distinct basis element b_z ; both sets have cardinality continuum. The scalar sum $\sum_i b_{z_i}$ uniquely specifies every multiplicity in the finite multiset, by rational linear independence. Thus one additive exact real, together with the known count, stores all empirical query information. This encoding is not a continuous finite-precision algorithm. Its existence shows why one cannot infer a lower bound from “additive” plus a nonlinear decoder alone. Arbitrary-real-weight mixtures defeat this example because real, rather than rational, linear relations are then allowed. Likewise, without any encoder regularity, the cardinality equality $|\mathbb{R}^D| = |\mathbb{R}|$ permits storing the whole moment vector in one ideal real. Even measurability alone is not a replacement for continuity: standard digit encodings can be made Borel measurable. This discontinuous-sum loophole is established prior theory: see Wagstaff et al., Theorem 3.2 and Supplement B.2[6].

If a decoder is required to be affine in the stored feature averages, the span criterion and D lower bound follow already from testing single records (or a dataset consisting of n copies of one record). This is a different, stronger restriction on decoding.

4 A finite-bit lower bound with explicit error metric

Let $\|A\|_{\max} = \max_{ij} |A_{ij}|$. Define simultaneous query error on the fixed finite family (4) by

$$e_z = \max_{(B,c) \in \mathcal{Q}} \|\widehat{G}_z(B, c) - G_z(B, c)\|_{\max}. \quad (6)$$

This metric is specified because an unqualified phrase such as “ ε -accurate residual queries” does not define a memory problem. All matrices F^{kl} have Frobenius norm at most one, so the query family uses bounded coefficients.

Lemma 5 (Stable recovery from the finite query family). *If $e_z \leq \varepsilon$, the formulas in the first theorem reconstruct a vector $\widehat{M}$ satisfying $\|\widehat{M} - M(z)\|_{\infty} \leq 12\varepsilon$.*

Proof. The errors in $S_y = G(0, 0)$, $\bar{H} = 2(G(0, 0) - G(0, 1))$, and $K_{ij,kl} = 2(G(0, 0) - G(F^{kl}, 0))_{ij}$ are at most ε , 4ε , and 4ε , respectively. Equation (2) has error at most 4ε . Equation (3) adds at most two such errors to the error of K , giving 12ε . To select one value for each fourth-order coordinate, fix any deterministic pair grouping in (3); approximate reconstructions need not otherwise respect redundant tensor identities. $\square$

Theorem 6 (Local worst-case bit bound). *Fix $d \geq 1$ and $n \geq D_0$. There exist a moment vector m_0 , $r > 0$, and a continuous section s on a neighborhood of $C = m_0 + [-r, r]^D$, with $F_n(s(m)) = m$. Fix the compact family of size- n unweighted datasets*

$$\mathcal{K} = s(C).$$

Any summary with at most 2^b states and any decoder satisfying (6) with $e_z \leq \varepsilon$ for every $z \in \mathcal{K}$ must obey, for $\varepsilon > 0$,

$$b \geq D \log_2 \left(1 + \left\lfloor \frac{2r}{25\varepsilon} \right\rfloor \right). \quad (7)$$

If S_y is supplied exactly as uncharged side information, the number of additional bits has the same lower bound with D_0 replacing D . No finite number of states can be exact on all of $\mathcal{K}$. These claims require correctness only on this compact family, not on all real-valued datasets.

Proof. The local section gives an open realizable moment set. Choose a closed cube C inside it; its continuous image $\mathcal{K} = s(C)$ is compact. In each coordinate place $L = 1 + \lfloor 2r/(25\varepsilon) \rfloor$ equally spaced grid values with spacing 25ε ; choose one realizing dataset via the section for each of its $N = L^D$ grid points. If two grid datasets had the same stored state, the decoder would return the same fixed query outputs for both. Recovering M from these outputs would approximate both moment vectors within 12ε , so their distance would be at most 24ε , contradicting the grid separation. Hence there are at least N states, proving (7).

For side information, fix the first p coordinates of the cube (the label moment) at their center, and vary only the remaining D_0 coordinates. All datasets now have identical side information, and the same proof applies. For zero error, even a line segment inside either cube has infinitely many distinct exact query vectors, so finite-state universal exact recovery is impossible. $\square$

The compact-family restriction makes a fixed positive absolute error meaningful. If one instead required the same guarantee over all unbounded real-valued datasets, already $G_z(0, 0) = S_y$ has unbounded range, so no finite-state code could have a finite uniform absolute error. Such a global impossibility would not be a useful quantitative precision bound.

This is a packing/pigeonhole theorem, not an invocation of Fano's inequality without a probability model. It has an optional precise randomized version. Draw the dataset uniformly from the constructed N -point packing. A randomized b -bit encoding/decoding scheme with shared randomness independent of the dataset and simultaneous success probability at least $1 - \delta$ on each dataset satisfies

$$b \geq \log_2 N + \log_2(1 - \delta). \quad (8)$$

Condition on all random coins: each state can be successful for at most one packed dataset, so at most 2^b of the N datasets succeed; average this inequality over the coins. This also covers private coins by revealing all of them as free shared randomness. A per-query success guarantee is different from simultaneous success for (4), and requires its own union-bound adjustment.

For fixed d, n and the resulting local scale r , (7) gives $D \log_2(1/\varepsilon) - O_{d,n,r}(1)$ as $\varepsilon \downarrow 0$. The proof does not give a dimension-uniform lower bound on r , a Gaussian-prior rate-distortion function, or a practical fixed-precision word count. The same compact family has a matching-order elementary

upper bound. For $\eta = \varepsilon/3$, use a grid on each canonical coordinate interval of length $2r$ with $J = 1 + \lceil 2r/\eta \rceil$ values and spacing at most η . Encoding the nearest grid vector has coordinate error at most η and uses no more than

$$b_{\text{upper}} = \left\lceil D \log_2 \left(1 + \left\lceil \frac{6r}{\varepsilon} \right\rceil \right) \right\rceil \quad (9)$$

bits. In (1)–(3), the errors in S_y , $\overline{H}$, and K are at most η , η , and 3η ; hence every query in (4) has error at most $3\eta \leq \varepsilon$. The grid vector need not itself be a realizable empirical moment vector for this decoder to work. If S_y is supplied exactly, quantize only the D_0 input moment coordinates and replace D by D_0 in (9). Thus on the specified compact family, the lower and upper bounds have matching $D \log_2(1/\varepsilon)$ leading order for fixed d, n, r .

In particular, neither the continuous exact theorem nor this local high-accuracy packing result proves that useful approximate residual queries require $\Omega(d^4)$ storage at a fixed practical tolerance. Classical covariance sketches such as Frequent Directions explicitly trade storage against covariance approximation error[7]; their applicability and bounds for the lifted quadratic features require a separate calculation. The local radius r in (7) must not be silently treated as dimension independent.

5 Concrete collisions and the Phase6 boundary

The missing fourth moment is operational. For $d = 1, n = 2$ and both labels zero, take dataset A with $x = (-1, 1)$ and dataset B with $x = (0, \sqrt{2})$. Both have $S_y = 0$ and $M_2 = 1$ (equivalently $\overline{H} = 0$). Dataset A has $\langle H^2 \rangle = 0$, whereas dataset B has $\langle H^2 \rangle = 1$. At $B = 1, c = 0$, their exact query values are 0 and $-1/2$. Thus even S_y together with the empirical second moment cannot answer all residual queries. Replicating each record equally gives the same collision at any even sample size.

Implementation inspected. The Phase6 source `code/frontier.py`, lines 358–365, updates

$$\text{moment} += \frac{1}{2} \sum_{\text{new } i} y_i (x_i x_i^\top - I), \quad \text{matrix} = \text{moment} / n - B.$$

Thus the proposal uses $S_y - B$. The exact historical empirical quantity is (1). For Gaussian inputs, $\mathbb{E}T(B) = B$ for fixed sample-independent B and $\mathbb{E}\overline{H} = 0$, but these are population identities. They do not turn $S_y - B$ into the realized empirical query. The Phase5 and Phase6 theory files already state this distinction correctly; this note supplies qualified minimality results that those files deliberately did not claim.

With the Phase6 value $d = 12$,

$$p = 78, \quad q = 1365, \quad D_0 = 1443, \quad D = 1521.$$

The sufficient sample-size threshold in the continuous lower bound is therefore $n \geq 1443$. With 1024 observations per arrival it applies from the second arrival if the query target is the whole historical empirical dataset. This observation is conditional on (U) and on what information the decoder can access. It is not a claim that the implemented approximate controller violates a memory constraint. A dense symmetric $p \times p$ Gram matrix plus $\overline{H}$ would use $p(p+1)/2 + p = 3159$ stored scalar slots, whereas the input polynomial moments use only 1443; adding S_y gives 3237 versus 1521. These compare representations, not bit lower bounds or runtime. Known constants and exact identities explain the redundancy.

Changing the requested output changes the target dimension. The matrix query $G(B, c)$ does not contain $\langle y \rangle$ or $\langle y^2 \rangle$. To reproduce the intercept gradient as well, store $\langle y \rangle$, one more independent coordinate:

$$\langle y - q_B - c \rangle = \langle y \rangle - \langle B, \bar{H} \rangle - c.$$

To reproduce the entire squared empirical loss, add $\langle y^2 \rangle$ too. The corresponding unrestricted moment dimensions are $D + 1$ and $D + 2$. Exact matrix gradient queries therefore do not alone reproduce factor Adam with its chosen minibatches, replay selection, coordinatewise states, and clipping.

Removing the intercept is dimension dependent. One must not simply subtract p coordinates when restricting queries to $c = 0$. For $d \geq 3$, the centered Gram moments already determine $\bar{H}$. The pointwise rank-one identities of $H + I$ imply

$$\begin{aligned} \langle H_{ii} + H_{jj} \rangle &= \langle H_{ij}^2 - H_{ii}H_{jj} \rangle - 1 & (i \neq j), \\ \langle H_{jk} \rangle &= \langle H_{ij}H_{ik} - H_{ii}H_{jk} \rangle & (i, j, k \text{ distinct}). \end{aligned}$$

Three pairwise diagonal sums recover each diagonal entry, and the second identity recovers each off-diagonal entry. Thus for $d \geq 3$ the dimension remains D even without c . For $d = 1$, the only input query coefficient is $\langle H^2 \rangle$, so the total affine moment dimension is two including S_y . For $d = 2$, the six distinct Gram polynomials are independent modulo constants: their degree-four leading terms force all coefficients to vanish except possibly a multiple of $H_{11}H_{22} - H_{12}^2$, and its degree-two part is nonzero. Hence the total is $6 + p = 9$, compared with $D = 11$ when all intercepts are queried. These are affine moment-space counts; the finite- n qualifications still apply.

6 Prior theory and defensible contribution

- **Classical least squares.** In the feature vector $h(x) = \text{vec}_{\text{Sym}}(xx^\top - I)$, equation (1) is the normal-equation residual $X^\top y - X^\top X\beta$, with the intercept cross term. MIT’s official notes give the ordinary normal equations and their gradient derivation[1]. The matrix query identity, Gram storage, and regression interpretation are established theory.
- **Statistical sufficiency is a different quantifier.** The official Berkeley notes define sufficiency through the conditional distribution of data given a statistic, relative to a specified statistical model, and minimal sufficiency through recovery from any sufficient statistic[2]. This note instead fixes a family of deterministic empirical queries. For a fixed design Gaussian regression model, the design is conditioned upon and its Gram matrix need not be charged as stored unknown information. This is another reason not to borrow statistical minimal-sufficiency claims as memory bounds here.
- **Polynomial moments and kernel means.** Sriperumbudur et al. study mean embeddings of measures and conditions for full distributional injectivity[3]. Taking the finite feature map to be our canonical polynomials makes M its empirical kernel mean. It distinguishes this query family but does not determine the full data distribution. The finite-dimensional feature interpretation, and the distinction between query equivalence and a characteristic kernel, are established theory.
- **Moment compression is classical.** Bayer and Teichmann prove a version of Tchakaloff’s theorem: finitely many weighted evaluation nodes can reproduce polynomial integrals through

a fixed degree[4]. This concerns existence of positive cubature representations, not a lower bound on physical memory. It reinforces that finite polynomial moment representations are old, and does not remove the coordinate or bit cost of describing nodes and weights.

- **Topology is classical.** The only topological input is Brouwer’s invariance of domain; Tao’s author-written exposition gives the theorem, its no-injection corollary, and a proof[5]. The dimension lower bound is its application to a locally full-rank empirical moment map.
- **Set-summary dimensional obstructions are prior work.** Wagstaff et al. make continuity versus discontinuous single-real encodings explicit and prove continuous latent-dimension obstructions for sum representations of set functions[6]. Their Theorem 4.1 proof uses an injection from an open ordered-set domain into the latent space. Our specialization concerns a fixed finite quadratic query family, and permits any continuous summary, including nonadditive and order-dependent ones. The dimension obstruction and the need to exclude pathological encodings are not new principles.

The defensible project contribution is the precise specialization: the $q + 2p$ count after polynomial dependencies and labels are charged; the $q + p$ additional cost given S_y ; a proof for sufficiently large fixed equal-weight samples rather than only arbitrary weighted measures; and the finite-query accuracy metric that yields a bit packing bound. The current search did not locate a paper stating this exact package for $H = xx^\top - I$, but this is not evidence of major theoretical novelty. It is best presented as a rigorous information boundary for the project’s historical summary, with classical ingredients credited. A substantial new algorithmic or learning-theoretic claim needs a separate result.

Inspection and validation. Read-only inputs were the Phase5 source-counterexample, nonlinear projection, and theory manuscripts; the Phase6 theory manuscript, `code/frontier.py`, and its dimension configuration. No original project files, training runs, or software installations were changed. An independent mathematical countercheck verified the continuous proof and its $n \geq D_0$ refinement. The companion pure-Python check verifies feature-space ranks for small dimensions, the explicit collision, and empirical query reconstruction using exact rational arithmetic.

References

- [1] MIT OpenCourseWare, *18.443 Statistics for Applications, Lecture 15: Regression Analysis*, Spring 2015, normal equations and normal linear regression slides. Official lecture PDF.
- [2] UC Berkeley, *Stat 210A Reader: Sufficiency*, Fall 2024. Official course reader.
- [3] B. K. Sriperumbudur, A. Gretton, K. Fukumizu, B. Schölkopf, G. R. G. Lanckriet, *Hilbert Space Embeddings and Metrics on Probability Measures*, JMLR 11(50):1517–1561, 2010. JMLR publication page.
- [4] C. Bayer and J. Teichmann, *The proof of Tchakaloff’s theorem*, Proc. Amer. Math. Soc. 134(10):3035–3040, 2006; preprint arXiv:math/0502473v2, 2005. Author preprint.
- [5] T. Tao, *Brouwer’s fixed point and invariance of domain theorems, and Hilbert’s fifth problem*, author exposition, 13 June 2011, Theorem 2 and Corollary 3. Author exposition and proof.

- [6] E. Wagstaff, F. B. Fuchs, M. Engelcke, I. Posner, M. A. Osborne, *On the Limitations of Representing Functions on Sets*, ICML, PMLR 97:6487–6494, 2019. Theorems 3.2 and 4.1, Supplement B.2. Official PMLR publication and supplements.
- [7] M. Ghashami, E. Liberty, J. M. Phillips, D. P. Woodruff, *Frequent Directions: Simple and Deterministic Matrix Sketching*, arXiv:1501.01711, 2015. Author preprint.